

Lab 15: Errors and Debugging

- Due Wednesday by 11:59pm
- Points 5
- Submitting an external tool
- Available Jul 6 at 5:59pm - Jul 8 at 11:59pm

Learning Objectives

- Practice writing tests for function.

Completion Requirements

- All function must be in Lab15.py
- Submit Lab15.py.
- Part of this lab will be manually graded.

Q1: Write Tests 1

You are given the specifications for the function, `find_first_valid_element(data, lower_bound, upper_bound, must_be_even)`. You do not need to write this function.

Function Specifications

Function Signature: `find_first_valid_element(data:list, lower_bound:int, upper_bound:int, must_be_even:boolean) -> int`

The function returns the index of the first integer element in the list, `data`, that is between `lower_bound` and `upper_bound` (both inclusive). If `must_be_even` is true, then index of the first even number within the range must be returned. If there is no element that matches the conditions then return -1. The list may contain non-integer elements. The search must be restricted to integer elements only.

Your Task

Add the following line at the top on Lab15.py.

```
from Lab15_provided import find_first_valid_element
```

More on how this works in the next lab

Write a function, `find_first_valid_element_test()`, that tests the `find_first_valid_element`. You must think of different cases that are possible for `find_first_valid_element` based on the specifications.

The lab videos are a helpful guide here. We will automatically score your code based on the number of cases that your tests covers. You must include checks for expected output in your program and print if the test passed or failed. The function output printed output will be graded manually.

Q2: Write Tests 2

You are given the specifications for the function, `scale_and_clamp_collection(items, multiplier, lower_limit, upper_limit)`. You do not need to write this function.

Function Specifications

Function Signature: `scale_and_clamp_collection(items:list, multiplier, lower_limit, upper_limit) -> None`

The function modifies the list, `items`, in-place by multiplying each numerical element by multiplier. If the resulting multiplied value is less than `lower_limit`, it is updated to `lower_limit`. If it is greater than `upper_limit`, it is updated to `upper_limit`. The function returns `None`. The list may contain non-numeric elements (or booleans), which must be ignored and left completely unchanged.

Your Task

Add the following line at the top on Lab15.py.

```
from Lab15_provided import scale_and_clamp_collection          # More on how this works in the next lab
```

Write a function, `scale_and_clamp_collection_test()`, that tests the `scale_and_clamp_collection`. You must think of different cases that are possible for `scale_and_clamp_collection` based on the specifications. The lab videos are a helpful guide here. We will automatically score your code based on the number of cases that your tests covers. You must include checks for expected output in your program and print if the test passed or failed. The function output printed output will be graded manually.

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